GRADUATE STUDY EDITION

Hybrid-Electric Bus System Modeling

A MATLAB and Simulink textbook for backward-facing energy analysis, dual-battery supervisory control, route studies, and configuration optimization

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Preface

This book explains a complete concept-level hybrid-electric bus study environment. It is written for graduate students who already know basic mechanics, electrical power, and MATLAB syntax, but who may be new to model-based design and energy-management architecture.

The project combines a transparent backward-facing physical model, an editable Simulink representation, an engineer-maintained Excel database, a programmatic MATLAB app, a bounded configuration optimizer, and automated verification. Its purpose is educational and architectural: it shows how equations, assumptions, data, software interfaces, and test evidence must remain consistent across a system model.

How to read this book
Chapters 1-3 establish the system and data architecture. Chapters 4-10 derive the physical and supervisory models. Chapters 11-15 explain execution, optimization, routes, and validation. Chapters 16-17 interpret worked studies and define responsible extensions.

Learning outcomes

- Derive tractive force and wheel power from a prescribed route speed and grade.
- Separate the traction DC-bus path from the genset-to-standby charging path.
- Explain a deterministic 30% dual-battery role-swap strategy and constant-point genset control.
- Derive the auxiliary-first, active-battery-second, resistor-bank-third regeneration allocation and verify its power balance.
- Distinguish energy accounting, equivalent replenishment, charge sustaining comparison, and true feasibility.
- Audit route provenance, component assumptions, numerical implementation, and validation evidence.
- Design extensions that increase fidelity without hiding uncertainty.

Scope statement

This is a concept energy model. It does not claim certification, manufacturer performance, thermal safety, battery ageing, emissions compliance, or real-time controller readiness. Synthetic component values are used deliberately so that the architecture can be studied without implying supplier data.

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Nomenclature and Conventions

Table F.1. Principal symbols and units

Symbol / term	Meaning	Typical unit
v, a	vehicle speed and longitudinal acceleration	m/s, m/s2
m	selected total vehicle mass	kg
F_{trac}	net tractive force required at the tyre contact patch	N
P_{wheel}	wheel mechanical power; positive traction, negative braking	kW
$P_{motor,dc}$	motor-side DC bus power; negative during regeneration	kW
P_{reg}	non-negative regenerative DC power available for allocation	kW
P_{dump}	surplus regenerative power dissipated by the resistor load bank	kW
SOE	stored usable battery energy divided by usable-energy rating	fraction or %
BSFC	brake-specific fuel consumption	g/kWh
η	directional efficiency	fraction
C_{rr}	rolling resistance coefficient	1
ρ	air density	kg/m3
Cd, A	drag coefficient and frontal area	1, m2

Sign convention
Battery power is positive when a pack discharges to the traction DC bus and negative while charging. Wheel power is positive for traction and negative for braking. Genset power is positive into the isolated standby charger; it never offsets traction demand.

1 Project Orientation

The project asks a systems question: for a prescribed bus mission, selected component set, loading state, environment, and supervisory calibration, how much wheel energy, battery energy, diesel fuel, grid-replenishment energy, operating cost, and feasible range result?

1.1 Why a backward-facing model?

A backward-facing model starts from the route speed that the vehicle is assumed to achieve. It differentiates speed to obtain acceleration, computes the tractive force required at the wheels, and then propagates power demand backward through the driveline to the DC bus and energy sources. This is efficient for energy sizing and comparison because it avoids solving a driver controller and vehicle-speed tracking loop.

Interpretation boundary
A backward model estimates the energy and power that would be required to follow the trace. It does not prove that the selected vehicle can track the trace. The project therefore logs unmet traction energy whenever component limits prevent delivery.

1.2 Project deliverables

Table 1.1. Primary project artifacts

Artifact	Role in the study
HybridBus_ComponentDatabase.xlsx	Master selections, catalogs, maps, routes, prices, and control calibrations
HybridBus_BackwardModel.slx	Editable ordinary-block Simulink representation with ten top-level subsystems
simulate_hybrid_bus_core.m	Detailed discrete reference kernel used by batch runs and optimization
HybridBusApp.m	Explorer app for selecting, simulating, plotting, optimizing, and exporting
tests/run_all_hybrid_bus_tests.m	Twenty-two assertion-based physics, isolation, regeneration-priority, control, cost, and ranking tests

1.3 Architecture at a glance

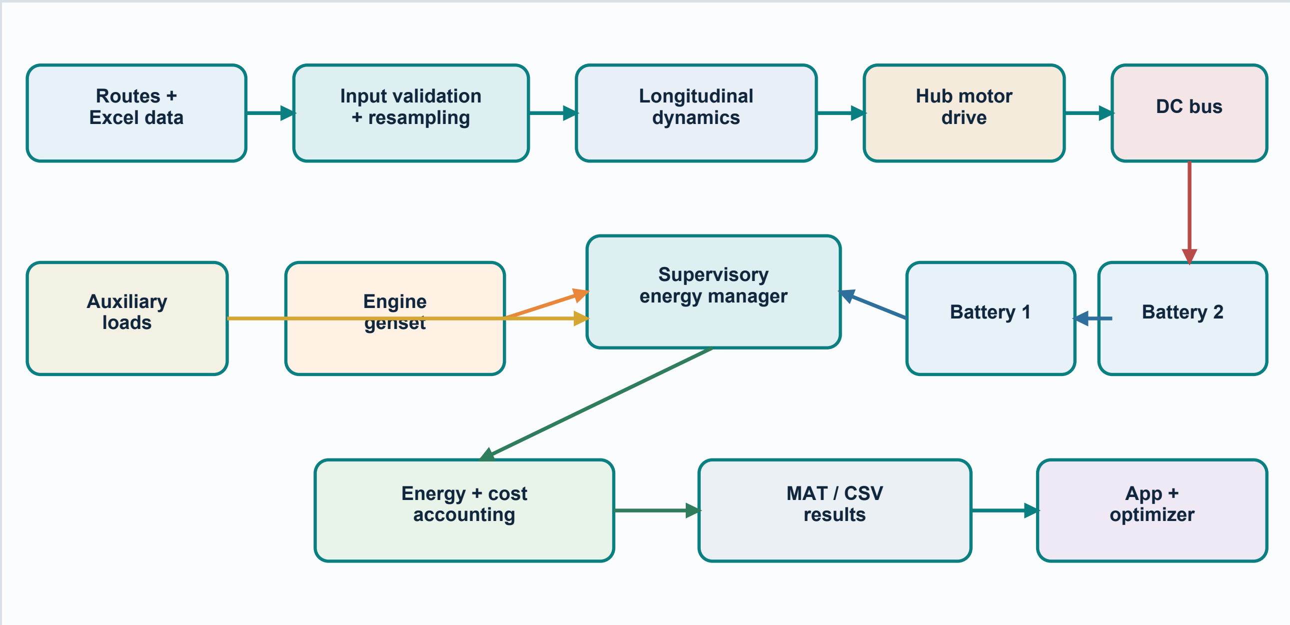

Figure 1.1. End-to-end project architecture from sourced routes to app and optimizer outputs

The Excel workbook is the single engineer-editable input source. The loader resolves stable IDs, the validator checks schema and physical ranges, and the input-preparation function resamples the selected route to a 1 s grid. Both the MATLAB kernel and the SLX model consume the same resolved parameter set.

2 Hybrid-Electric System Architecture

2.1 Energy topology

The architecture contains two independent battery packs, two rear hub motors, a diesel engine-generator set, an auxiliary electrical load, and a resistor load bank. The topology has two deliberately separated electrical paths. Only the active battery connects to the traction DC bus; the genset connects only to the standby battery through a dedicated charger. The genset therefore cannot propel the vehicle or reduce active-battery traction demand. During braking, regenerated electrical power supplies auxiliaries first, charges only the active battery second, and is sent to the resistor load bank if the active battery cannot accept the remainder.

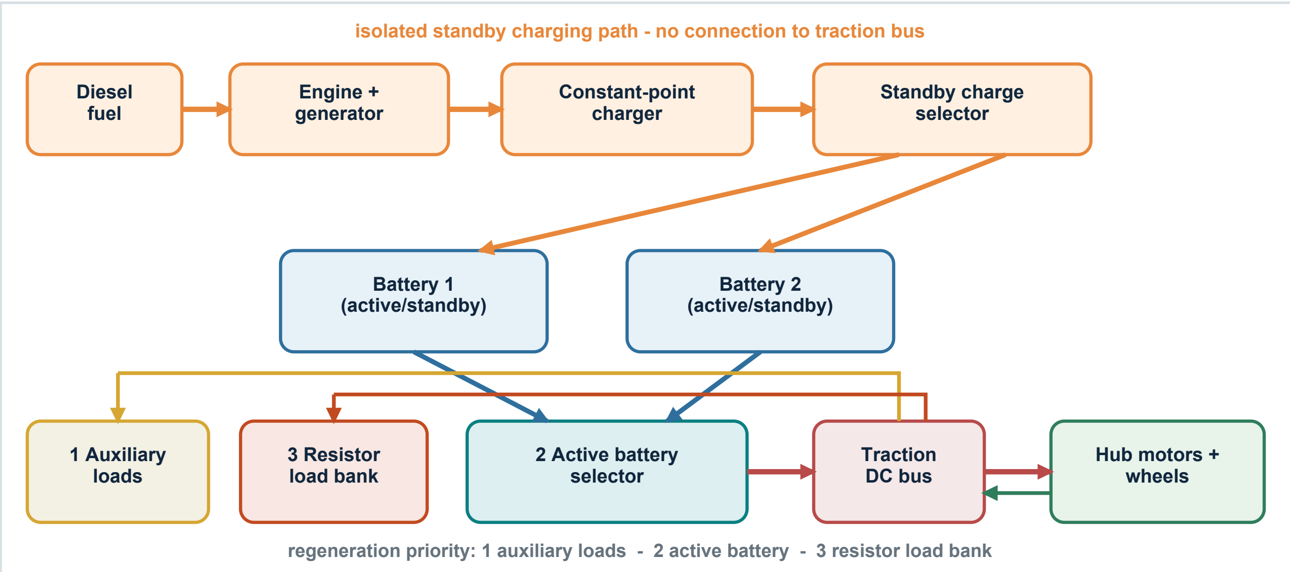

Figure 2.1. Conceptual energy flow, isolated genset path, and regeneration priorities

2.2 Simulink decomposition

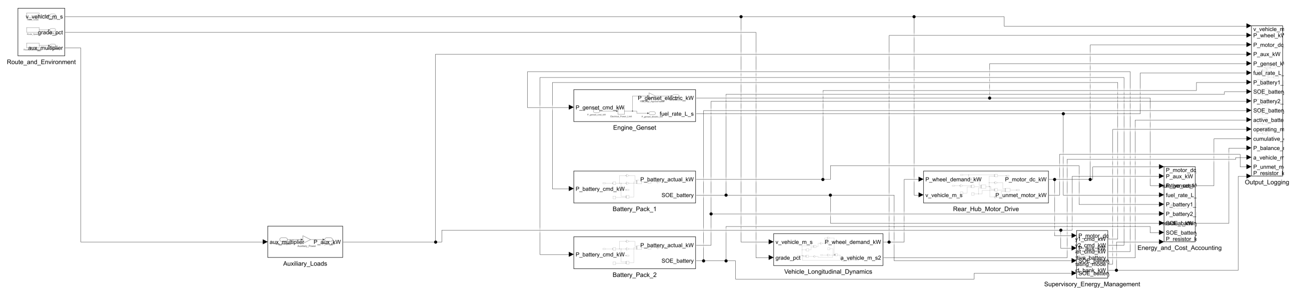

Figure 2.2. Top-level Simulink model and logged signal fan-out

Table 2.1. Top-level model responsibilities

Subsystem	Responsibility
Route_and_Environment	Publishes route speed, grade, and auxiliary multiplier
Vehicle_Longitudinal_Dynamics	Computes filtered acceleration and required wheel power
Rear_Hub_Motor_Drive	Applies motor torque, power, speed, efficiency, and regeneration limits
Auxiliary_Loads	Calculates base and HVAC electrical demand
Supervisory_Energy_Management	Enforces the 30% role swap, routes fixed genset power only to standby, and applies auxiliary-active-dump regeneration priority
Battery_Pack_1 / Battery_Pack_2	Integrate independent energy states under power and SOE limits
Engine_Genset	Produces constant optimum electrical power when enabled and calculates fuel rate
Energy_and_Cost_Accounting	Includes resistor-bank dissipation in the DC balance and accumulates energy/cost quantities
Output_Logging	Publishes stable workspace signals for inspection

2.3 Reference kernel versus SLX

The ordinary-block SLX is intended for architecture inspection, signal tracing, and interactive simulation. The MATLAB kernel contains the complete deterministic limit handling and supervisory behavior used by batch studies and the optimizer. This separation keeps the teaching model editable while making the numerical reference path easy to test.

3 Data Architecture and Reproducible Inputs

3.1 Workbook as engineering database

The version 1.4.0 workbook contains 23 sheets. IDs rather than row numbers form the interfaces between the Dashboard, catalogs, input resolver, app, and optimizer. This is important: a stable ID survives sorting and insertion, while a row number does not.

Table 3.1. Workbook organization

Sheet group	Examples	Purpose
Selections	Dashboard	Chosen component IDs, SOEs, prices, fuel tank, and controls
Component catalogs	Battery, Motor, Genset, Tyre, Final Drive, Mass	Ratings, efficiency, compatibility, and enable flags
Maps	Engine_Fuel_Map, Generator_Efficiency_Map	Bounded lookup data for fuel and conversion efficiency
Routes	Route_Catalog, Route_Time_Speed, Route_Distance_Speed, Route_Grade	Provenance and mission traces
Calibration	Environment, Control_Calibration, Energy_Prices	Scenario and supervisor parameters
Governance	Units_and_Definitions, Change_Log	Definitions and database history

3.2 Input pipeline

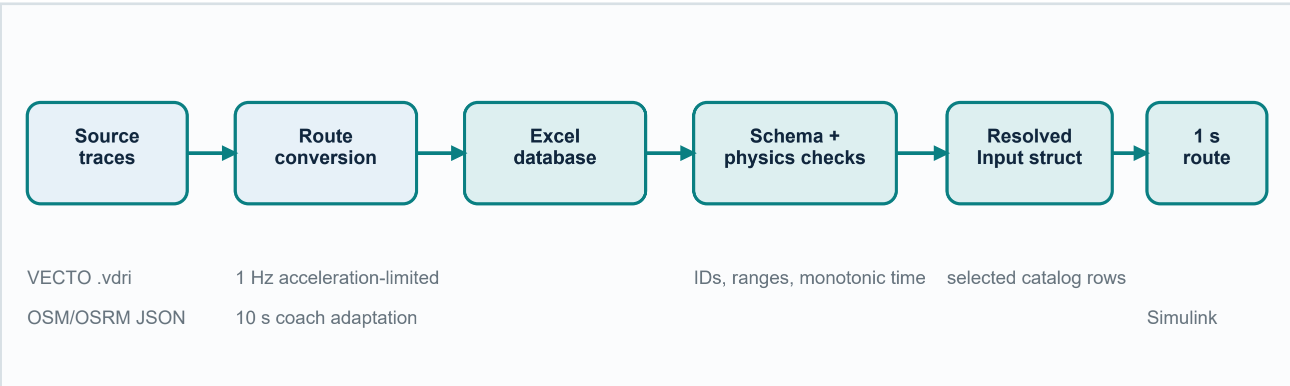

Figure 3.1. Route and component data pipeline

The loader imports every sheet with preserved column names. Validation then checks required sheets, unique IDs, battery and motor consistency, total-mass positivity, route finiteness, monotonic time, non-negative speed, acceleration bounds, and nonzero distance. Preparation resolves selected rows into scalar structures and resamples speed, grade, and auxiliary multiplier at the model sample time.

3.3 Default configuration

Table 3.2. Default case resolved from the Dashboard

Input	Selection / value	Unit or interpretation
Route	VECTO-URBAN	stable route ID
Battery 1 / 2	BAT-12 / BAT-12	independent packs
Motor / genset	MOT-12 / GEN-05	two identical hub motors + one genset
Total mass	19,000	kg
Initial B1 / B2 SOE	85 / 75	%
Fuel / electricity price	1.65 / 0.18	EUR/L / EUR/kWh

4 Longitudinal Vehicle Dynamics

4.1 Speed, acceleration, and grade

The route speed is prescribed. A first-order discrete filter is applied to the raw finite-difference acceleration to reduce numerical spikes without introducing a continuous state.

$$a_{raw,k} = (v_k - v_{k-1}) / \Delta t_{k-1} \tag{4.1}$$

$$a_k = a_{k-1} + [\Delta t / (\tau + \Delta t)] (a_{raw,k} - a_{k-1}) \tag{4.2}$$

Road grade is stored as percent. The model converts it to an angle using $\theta = \mathrm{atan}(grade/100)$. Headwind is added to vehicle speed and clipped so aerodynamic relative speed cannot become negative.

4.2 Road-load forces

$$F_{inertia} = m\, a \tag{4.3}$$

$$F_{roll} = m\, g\, C_{rr} \cos(\theta) \tag{4.4}$$

$$F_{grade} = m\, g \sin(\theta) \tag{4.5}$$

$$F_{aero} = 0.5\, \rho\, C_d\, A\, v_{air}^2 \tag{4.6}$$

$$F_{tractive} = F_{inertia} + F_{roll} + F_{grade} + F_{aero} \tag{4.7}$$

$$P_{wheel,demand} = F_{tractive}\, v / 1000 \tag{4.8}$$

Mass definition
The selected TotalVehicleMass_kg is the complete simulated mass. It already represents curb mass, passengers, cargo, batteries, motors, final drives, and genset. Component mass is reported separately for traceability and is not added a second time.

4.3 Force scaling intuition

Inertia, rolling resistance, and grade forces scale linearly with total mass. Aerodynamic force does not. Therefore mass strongly changes stop-and-go and climbing demand, while high-speed cruise also depends heavily on frontal area and drag coefficient.

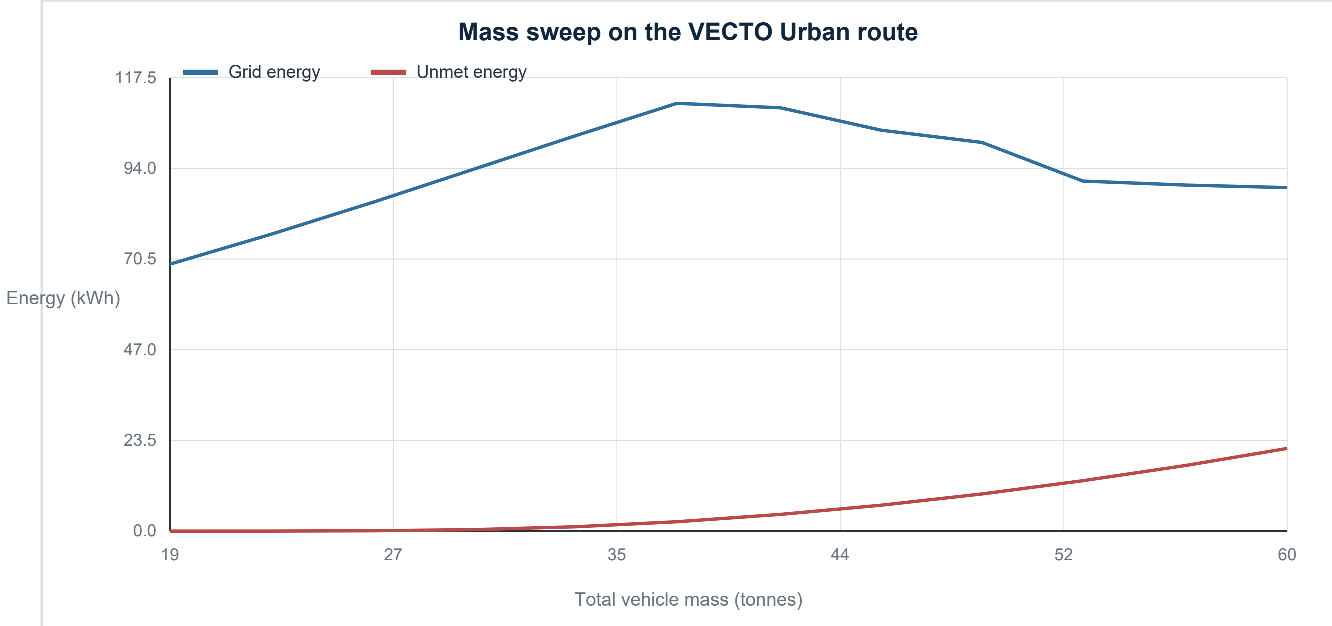

Figure 4.1. Energy and feasibility sensitivity from 19 to 60 tonnes

The default component set remains essentially feasible at the low end, but unmet traction energy grows with mass. Cost per kilometre alone must not be used to rank infeasible cases because unsupplied energy is not purchased.

5 Rear Hub Motors and Fixed Driveline

5.1 Kinematics

$$\omega_{wheel} = v / r_{loaded} \quad (5.1)$$

$$\omega_{motor} = i_{fd}\, \omega_{wheel},\; n_{motor} = 60\, \omega_{motor} / (2\pi) \quad (5.2)$$

The model uses two identical rear hub motors. The selected fixed reduction ratio connects wheel speed to motor speed. If motor speed exceeds the catalog maximum, available motor power is set to zero and the shortfall is logged.

5.2 Torque and power envelope

$$T_{available} = \min[T_{peak}, P_{peak} / \max(\omega_{motor}, \varepsilon)] \quad (5.3)$$

$$P_{tractive,max} = 2 \min(P_{peak}, T_{available}\, \omega_{motor})\, \eta_{fd,mot} \quad (5.4)$$

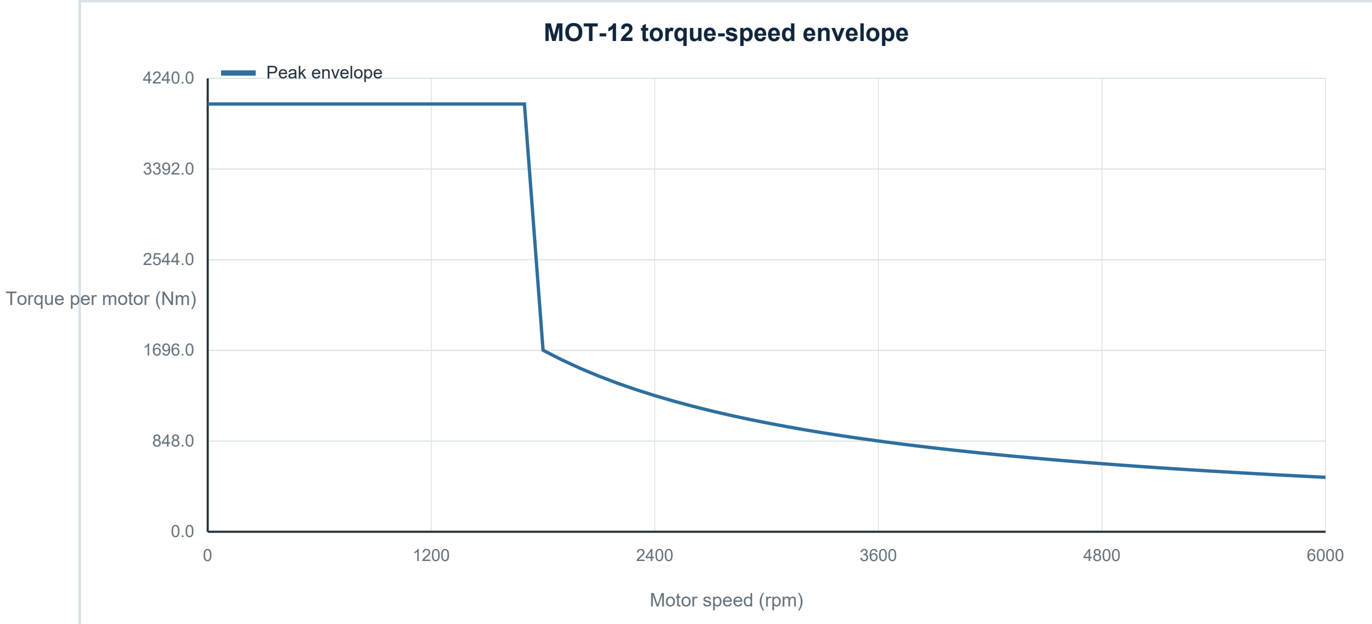

Figure 5.1. Selected motor constant-torque and constant-power regions

5.3 Directional efficiency and regeneration

Traction wheel power is divided by motor and final-drive motoring efficiencies to obtain DC demand. During braking, delivered negative wheel power is multiplied by regeneration efficiencies, producing negative motor DC power. Regeneration is limited by motor capability, fixed-drive direction, and the lower of the two pack regeneration ratings.

$$P_{motor,dc} = P_{wheel} / (\eta_{fd,mot}\, \eta_{motor,mot}) \text{ for traction} \quad (5.5)$$

$$P_{motor,dc} = P_{wheel}\, \eta_{fd,reg}\, \eta_{motor,reg} \text{ for regeneration} \quad (5.6)$$

6 Battery Energy Model

6.1 Energy state and SOE

Each battery is represented by usable stored energy rather than voltage, current, temperature, or electrochemical states. This is appropriate for concept energy analysis but cannot predict electrical transients or thermal derating.

$$SOE = E / E_{usable} \tag{6.1}$$

$$E_{k+1} = E_k - P_{dis}\, \Delta t / (3600\, \eta_{dis}) \tag{6.2}$$

$$E_{k+1} = E_k + |P_{chg}|\, \eta_{chg}\, \Delta t / 3600 \tag{6.3}$$

6.2 Three simultaneous limits

At every time step the battery helper applies the minimum of the requested power, catalog power limit including derating, and energy-implied power needed to remain within minimum or maximum SOE. Charge power is additionally bounded by the pack regeneration rating.

Numerical safeguard
The energy state is clamped to [MinSOE, MaxSOE] after each update. Regenerative power rejected by the active battery is assigned to the resistor load bank; a genset-charger mismatch remains a separately logged rejected-charge term. Unserved positive demand is recorded as unmet DC power.

6.3 Why one active traction pack?

The supervisor permits only one battery to discharge for traction. This is intentionally conservative: the active pack must be able to support residual DC demand by itself. The rule also creates a clean architecture for switching, standby charging, and deterministic regenerative routing.

Table 6.1. Synthetic battery catalog coverage

Catalog range	Minimum	Maximum
Usable energy	100 kWh	200 kWh
Maximum discharge	300 kW	500 kW
Maximum charge	150 kW	280 kW
Mass	850 kg	1450 kg

7 Engine-Generator Set and Fuel Model

7.1 Constant best-efficiency operating point

The genset has one permitted electrical command while on: the catalog OptimumPower_kW value, bounded by the rated maximum. It is not load-following and is not connected to the traction bus. Its dedicated charger sends power only to the pack currently designated standby. Battery charge-power and upper-SOE limits determine accepted power; any transient charger mismatch is recorded separately from regenerative load-bank dissipation.

7.2 Generator and engine maps

$$P_{mech} = P_{gen,dc} / \eta_{gen}(\lambda) \tag{7.1}$$

$$\dot{m}_{fuel} = \max[\dot{m}_{idle}, P_{mech}\ BSFC(\lambda) / (1000\ \rho_{fuel}\ 3600)] \tag{7.2}$$

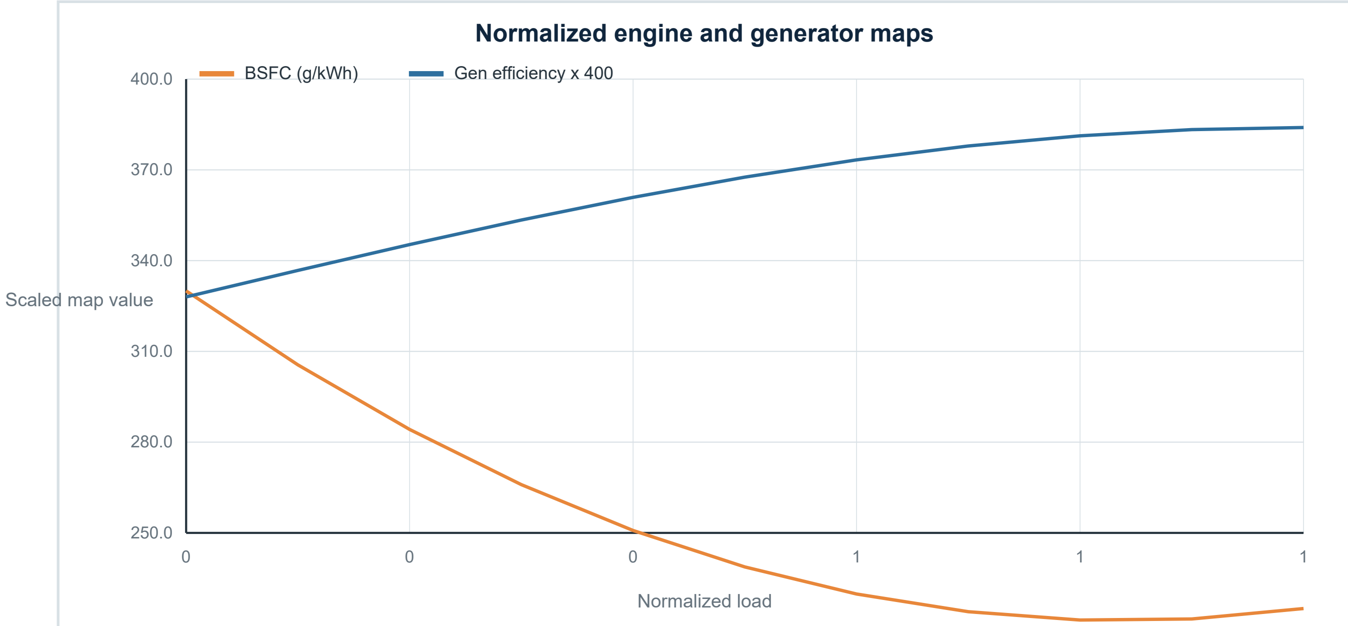

Figure 7.1. Synthetic BSFC and generator-efficiency maps used for interpolation

The generator efficiency is clipped to a physically reasonable numerical range, and BSFC is linearly interpolated over normalized load. A fixed start-fuel penalty is added on every off-to-on transition.

7.3 What the fuel model does not claim

Not an emissions model
There is no engine warm-up, transient combustion, aftertreatment, ambient correction, fuel-temperature effect, or pollutant calculation. The output is a concept diesel-volume estimate, not an emissions or homologation result.

8 Supervisory Energy Management

8.1 Control objectives

The deterministic supervisor has five priorities: keep the genset electrically isolated from traction; preserve exactly one active traction pack; swap roles at 30% SOE when the alternate pack is ready; recharge the depleted standby pack at constant best-efficiency genset power; and allocate regeneration strictly to auxiliaries, then the active battery, then the resistor load bank.

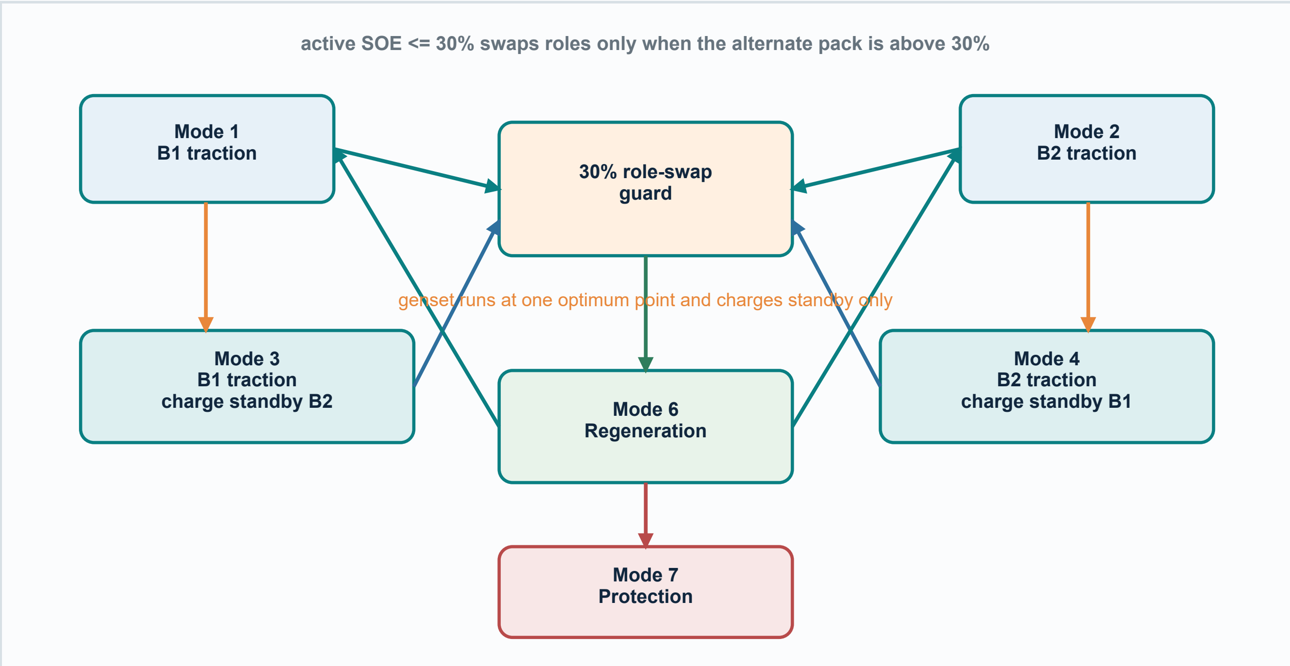

Figure 8.1. Supervisory operating modes and principal transitions

8.2 Decision order

1. Read both pack SOEs and retain the current active-battery state.
2. If active SOE is at or below 30% and the standby pack is above 30%, exchange the active and standby roles immediately.
3. Start the genset when standby SOE is at or below 30%; while on, command the constant OptimumPower_kW value.
4. Send positive motor-plus-auxiliary demand only to the active battery; the genset is excluded from this equation.
5. Send genset power only to the standby battery; the genset never participates in the traction-node balance.
6. For negative motor DC power, supply auxiliary demand first, send the remaining regenerative power only to the active battery, and divert any unaccepted remainder to the resistor load bank.
7. Update energy states, fuel, operating mode, losses, load-bank heat, rejected genset charge, and the balance residual.

8.3 Regeneration allocation by first principles

Let regenerative power be a non-negative available quantity. Auxiliary loads remain powered during braking, so they consume the first portion directly at the traction DC bus. Only the remainder is offered to the active pack. The battery helper then applies its instantaneous charge-power, regeneration-power, efficiency, and upper-SOE limits. The resistor load bank receives exactly the terminal power that the active battery cannot accept. The standby pack is excluded from this path because its only charging source is the isolated genset charger.

$$P_{reg} = \max(0, -P_{motor,dc}) \tag{8.1}$$

$$P_{reg,aux} = \min(P_{reg}, P_{aux}) \tag{8.2}$$

$$P_{reg,rem} = \max(0, P_{reg} - P_{reg,aux}) \tag{8.3}$$

$$P_{dump} = \max(0, P_{reg,rem} - P_{reg,active}) \tag{8.4}$$

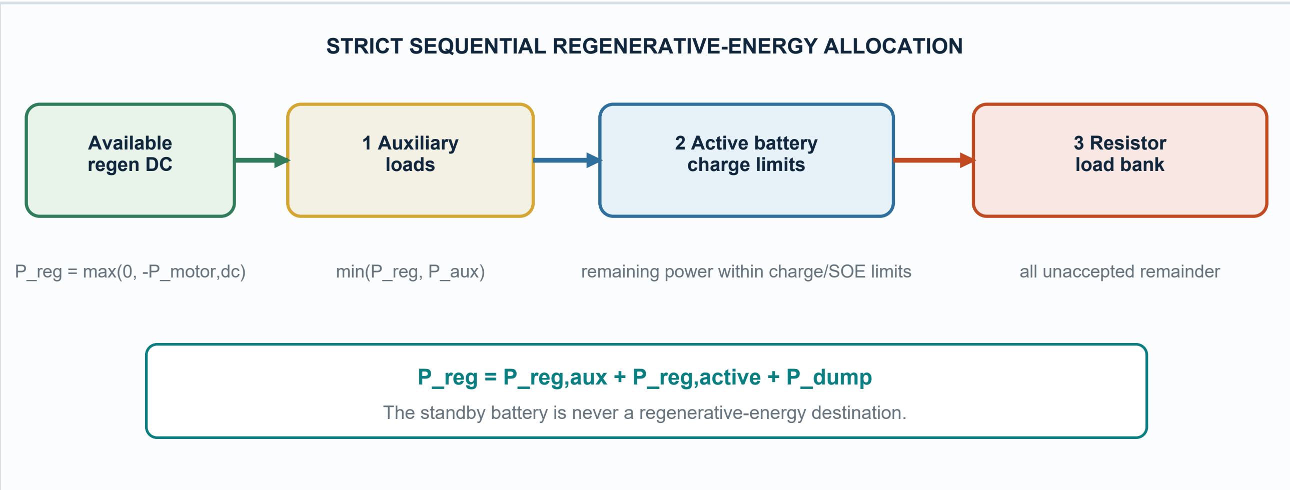

Figure 8.2. Strict auxiliary-active-battery-resistor regeneration hierarchy

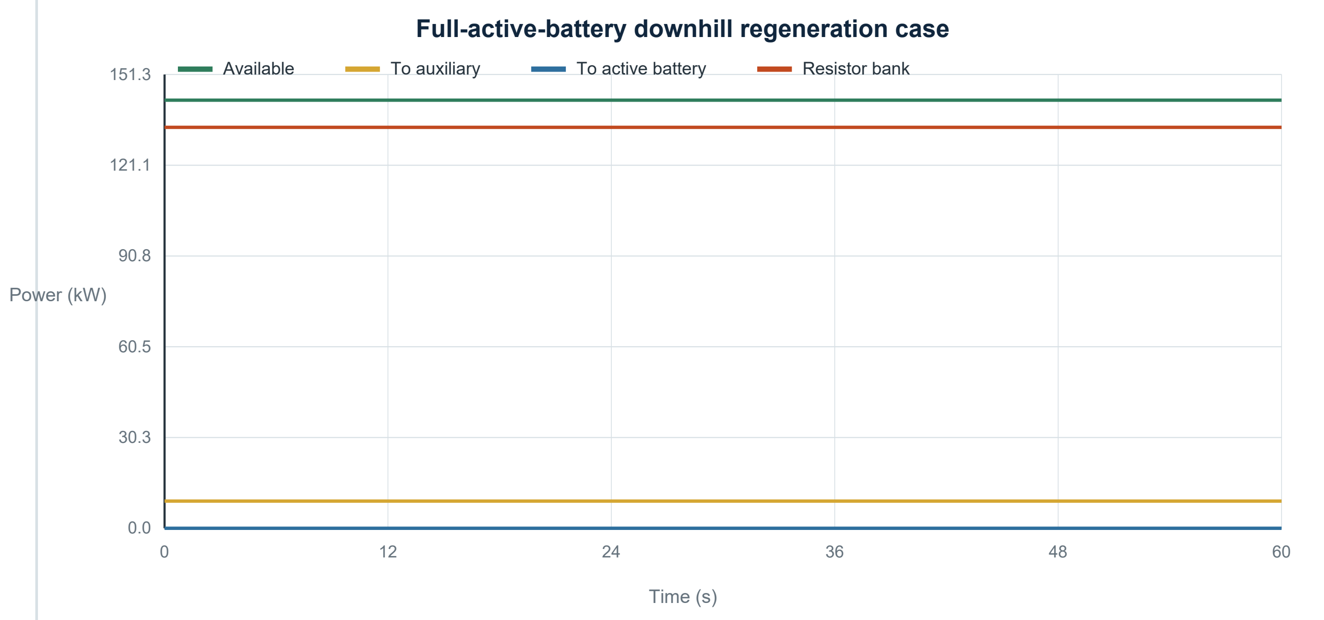

Figure 8.3. When the active battery is full, surplus regenerative power becomes controlled waste heat

Thermal interpretation
The resistor-bank signal is an electrical power sink. A production design would require continuous and transient thermal ratings, cooling airflow, temperature sensing, contactor diagnostics, insulation monitoring, and a braking fallback if the bank is unavailable. Those thermal and safety dynamics are outside the present concept model.

8.4 Hysteresis and dwell time

The 30% role threshold is an architectural rule rather than a tunable calibration. A role swap is allowed only if the alternate pack is above 30%; this prevents rapid swapping when both packs are depleted. If both are low, the current active pack remains connected while the genset charges standby. Charging continues until the standby pack reaches its upper SOE limit or becomes the active pack.

Table 8.1. Operating-mode interpretation

Mode	Meaning	Primary power behavior
1	Battery 1 active	B1 alone supplies positive traction-bus demand
2	Battery 2 active	B2 alone supplies positive traction-bus demand
3	B1 active; B2 charged	Genset fixed output is routed only to standby B2
4	B2 active; B1 charged	Genset fixed output is routed only to standby B1
6	Regeneration	Auxiliary first, active battery second, resistor load bank third
7	Protection	Sources are clamped and unmet demand is recorded

9 Power, Energy, Cost, and Range Accounting

9.1 DC power conservation

Two node equations make the isolation explicit. At the traction node, only the active battery balances motor power, auxiliary demand, and resistor-bank dissipation. At the charger node, the genset balances standby-battery charging and any transient rejected charger power. Summing the nodes gives the project-wide residual without implying a physical genset-to-traction connection.

$$P_{active} - P_{motor,dc} - P_{aux} + P_{unmet} - P_{dump} = 0 \quad (9.1a)$$

$$P_{gen} + P_{standby} - P_{gen,rejected} = 0 \quad (9.1b)$$

$$P_{gen} + P_{B1} + P_{B2} - P_{motor,dc} - P_{aux} + P_{unmet} - P_{dump} - P_{gen,rejected} = P_{residual} \quad (9.1c)$$

$$P_{rejected} = P_{dump} + P_{gen,rejected} \quad (9.1d)$$

9.2 Equivalent grid replenishment

$$E_{grid} = [\max(0,E_{1,0}-E_{1,f}) + \max(0,E_{2,0}-E_{2,f})] / \eta_{grid} \quad (9.2)$$

$$C_{total} = V_{fuel}\, c_{fuel} + E_{grid}\, c_{electricity} \quad (9.3)$$

$$C_{km} = C_{total} / d_{route} \quad (9.4)$$

Terminal battery surplus is not credited by default. This prevents an artificial negative electric cost and makes the equivalent-replenishment comparison conservative.

9.3 Range estimates

$$R_{battery} = E_{usable,initial} / e_{DC,route} \quad (9.5)$$

$$R_{fuel} = V_{tank} / v_{fuel,per\text{-}km} \quad (9.6)$$

Range is conditional
Range estimates assume that the observed route energy intensity and supervisory behavior remain representative. If the example route never starts the genset, the code estimates supported fuel range at the selected optimum genset point rather than reporting infinity.

9.4 Feasibility before economics

A low cost per kilometre is meaningful only when unmet traction energy is negligible and the balance residual satisfies tolerance. An infeasible case can look artificially cheap because energy that the vehicle failed to deliver was never purchased. The optimizer therefore filters feasibility before ranking cost.

10 Numerical Implementation in MATLAB and Simulink

10.1 Discrete solution

The default solver interpretation is fixed-step discrete with a 1 s sample time. Route speed, grade, and auxiliary multiplier are linearly interpolated; stop flags use previous-value interpolation. Energy updates use time-step integration in kWh, while fuel rate is integrated in litres.

10.2 Execution sequence

```
cd('C:\TempData\Hybrid_Vehicle\HybridBusProject')
project = startup_hybrid_bus;
Results = run_hybrid_bus_simulation;
open_system('HybridBus_BackwardModel.slx')
app = HybridBusApp;
```

The runner loads and validates the database, resolves the selected input structure, executes the detailed kernel, attaches validation metadata, and optionally exports MAT and CSV files. The SLX workspace publisher assigns only the stable, unit-suffixed variables needed by the block diagram.

10.3 Results structure

Table 10.1. Simulation output contract

Field	Contents
Metadata	database file/version, MATLAB release, model version, timestamp
SelectedConfiguration	stable IDs for route and every selected component/calibration
InputParameters	fully resolved scalar structures, maps, and resampled route
Signals	grouped vehicle, wheel, motor, auxiliary, regeneration, battery, genset, controller, and energy histories
Summary	distance, fuel, grid energy, regeneration allocation, load-bank heat, costs, range, SOE, unmet energy, residual, and mass
Validation	warnings, constraint violations, and feasibility flag

11 Using the Explorer App

11.1 Configuration panel

The programmatic UIFigure app presents route, battery, motor, genset, total-mass, and auxiliary selections in a fixed left sidebar. Initial battery SOEs are entered as percentages and converted to fractions only at the model boundary. Fuel and electricity price units are explicit.

The Powertrain Architecture tab shows the isolated standby-charging chain, active-battery selector, traction DC bus, regenerative return, auxiliary-first branch, and resistor load bank. It is the quickest visual check that genset power has no direct route to the motors or auxiliaries and that the standby battery is not a regenerative-energy destination.

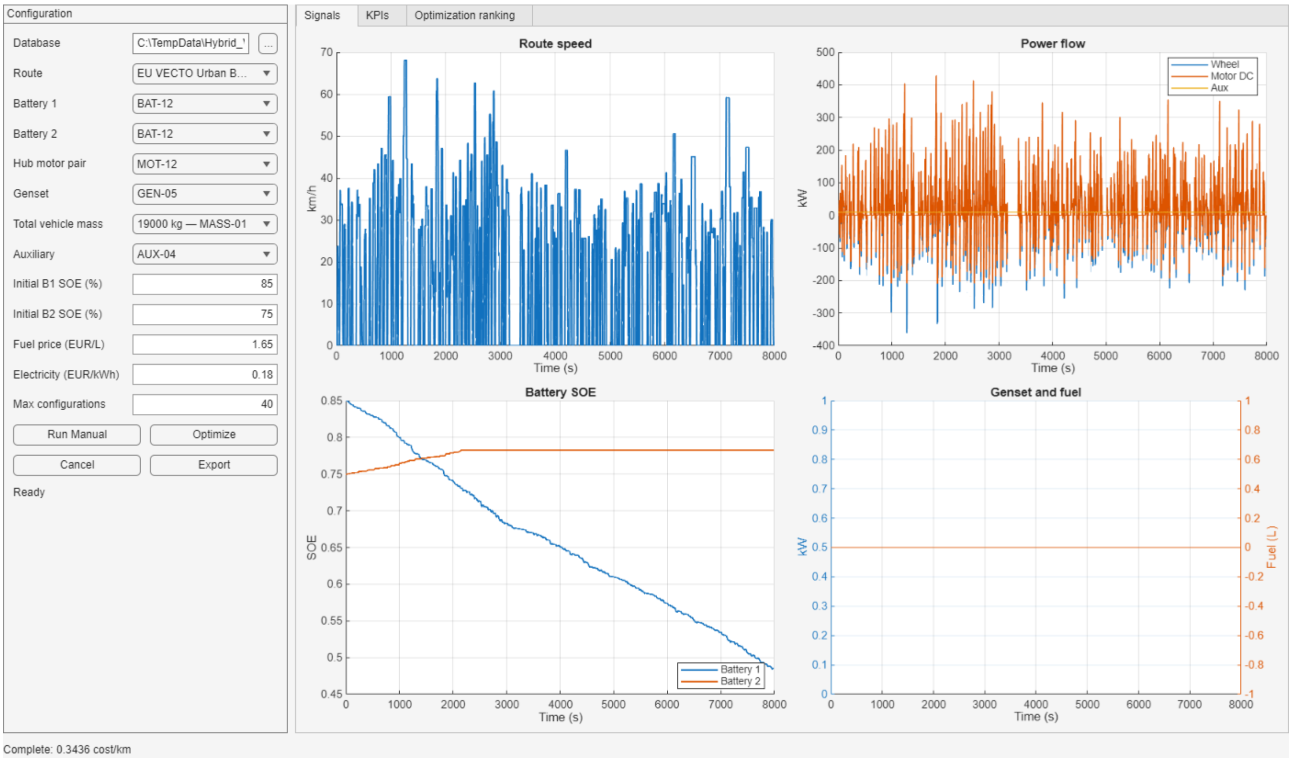

Figure 11.1. Explorer app after running the default VECTO Urban case

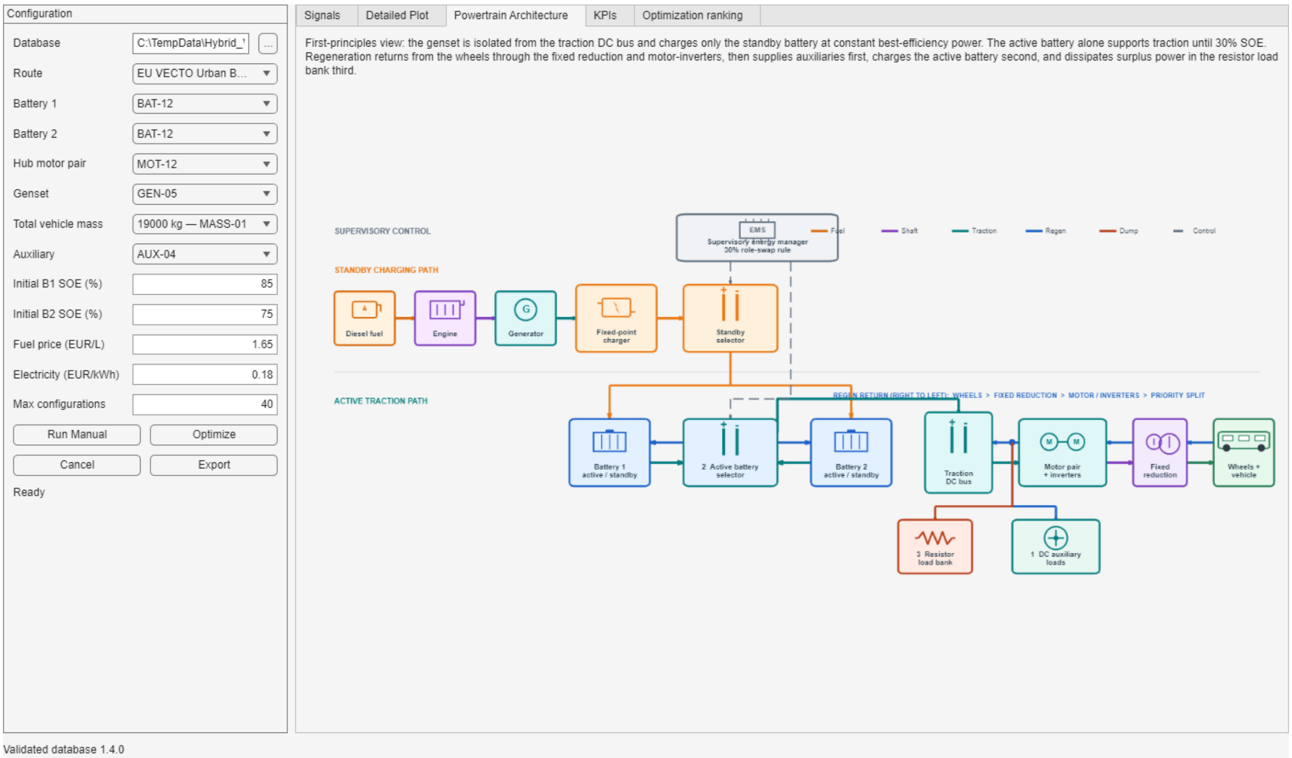

Figure 11.2. Architecture tab with the wheel-to-inverter regenerative return and numbered destination priorities

11.2 Manual-study workflow

1. Select the database and confirm that the status reports a validated version.
2. Choose a route, component set, total vehicle mass, initial SOEs, and prices.
3. Run the manual case and inspect speed, power flow, battery SOE, genset/fuel, KPIs, and warnings.
4. Treat any unmet traction energy as a failed design case, not as an acceptable low-cost solution.
5. Export the selected result only after checking assumptions and feasibility.

11.3 Optimization tab

The optimization view ranks feasible configurations and keeps stable component IDs visible. The app holds the selected mission and economic assumptions fixed while it varies Battery 1, the motor pair, genset, and final-drive choice. Battery 2 remains at the sidebar selection. Cancellation is checked between simulations, so the bounded search remains responsive without requiring Parallel Computing Toolbox. Chapter 12 defines the complete search and interpretation workflow.

12 Configuration Optimization

12.1 Purpose and study boundary

The optimizer is a deterministic design-screening tool. It answers: among the component combinations actually evaluated for one prescribed mission, which feasible combination has the lowest modeled operating cost per kilometre? It does not redesign components, tune continuous parameters, predict hardware life, or prove a global optimum.

Table 12.1. Study variables controlled by the Optimize button

Held fixed by the app	Varied by the app
Route and its speed, grade, dwell, and auxiliary multipliers	Battery 1 catalog selection
Total vehicle mass variant	Rear hub-motor pair
Battery 2 catalog selection	Engine-generator set
Initial Battery 1 and Battery 2 SOE	Final-drive ratio
Auxiliary-load profile and environment	
Fuel price and electricity price	

Why Battery 2 stays fixed
The app deliberately uses the current Battery 2 selection while varying Battery 1. This supports asymmetric active/standby studies and keeps the search bounded. The underlying optimizer function can vary Battery 2 when called programmatically with a different Vary list.

12.2 Candidate set and mixed-radix enumeration

For every varied category, the optimizer reads stable component IDs from catalog rows marked OptimizationEnabled. If the four candidate counts are nB, nM, nG, and nF, the complete discrete search space is

$$N_{space} = n_B \, n_M \, n_G \, n_F \tag{12.1}$$

A mixed-radix counter converts each linear evaluation index into one Battery 1, motor, genset, and final-drive combination. This avoids materializing a large Cartesian-product table in memory and gives repeatable ordering between runs with the same database.

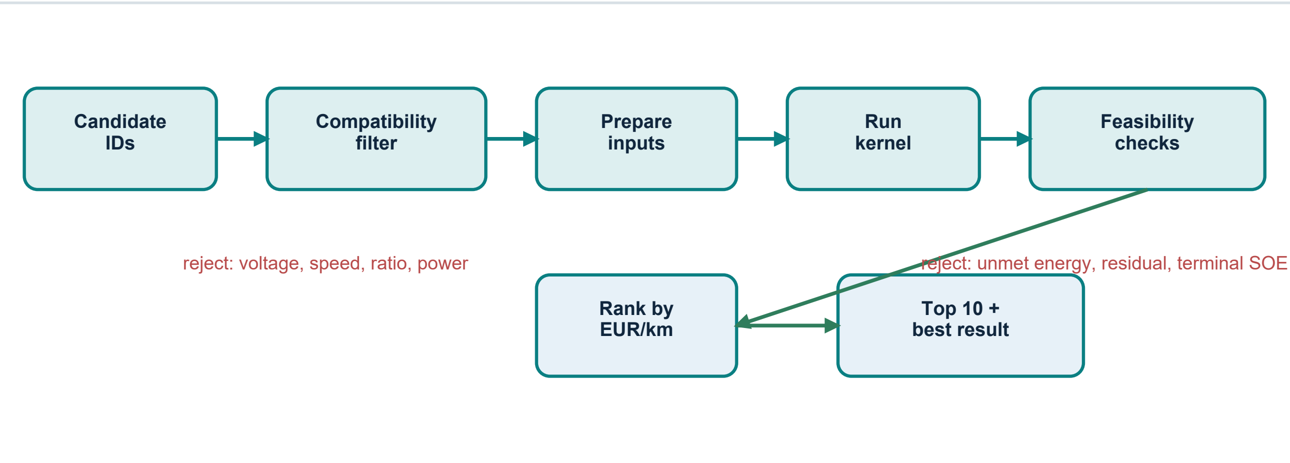

Figure 12.1. Compatibility, simulation, feasibility, and ranking sequence

12.3 Meaning of Max configurations

Max configurations is the evaluation budget for one click of Optimize. Its default value of 40 means that no more than forty candidate combinations are inspected in that run. Both compatible simulated cases and combinations rejected by the pre-simulation compatibility filter consume an evaluation number. The field is therefore a runtime-control setting, not a request for forty feasible solutions.

$$N_{evaluated} = \min(N_{max}, N_{space}) \tag{12.2}$$

Increasing the budget explores more of the deterministic catalog order and may find a lower-cost feasible design, but simulation time rises approximately with the number of compatible cases. If Nmax is smaller than Nspace, the reported winner is only the best among the evaluated prefix. A global discrete optimum is established only when the complete enabled search space is evaluated, or when a separately justified search strategy covers it.

Table 12.2. How to interpret the evaluation budget

Setting	Interpretation	Expected effect
Small budget	Rapid screening of the first enabled combinations	Fast, but stronger risk of missing a better design
40 (default)	At most forty candidates are checked	Practical interactive study size
At least Nspace	Every enabled combination is checked	Exhaustive discrete ranking for the stated catalogs

12.4 Compatibility filters

Compatibility filtering removes combinations that violate static interface or rating constraints before the drive-cycle simulation. Rejected rows remain in the evaluated-results table with a reason, making the search auditable.

- Battery-to-motor DC voltage class difference no greater than 50 V.
- Motor speed below maximum at the selected route speed, tyre radius, and final-drive ratio.
- Final-drive ratio inside the selected motor compatibility interval.
- At least one selected battery has adequate continuous discharge rating for the motor rating.
- Genset optimum power is within its rating and within either standby battery's charge-power capability.

12.5 Simulation and feasibility gates

Every compatible row is resolved into a complete model input and simulated over the selected route. A low cost is not sufficient. The row enters the feasible ranking only if it passes all three gates below.

$$E_{unmet,traction} < 10^{-3}\ \mathrm{kWh} \tag{12.3}$$

$$E_{balance,error} \le E_{balance,tolerance} \tag{12.4}$$

$$|SOE_{combined,final} - SOE_{combined,target}| \le \Delta SOE_{terminal} \tag{12.5}$$

The first gate rejects component sets that cannot satisfy requested wheel demand. The second detects inconsistent source, sink, or loss accounting. The third prevents a configuration from appearing economical merely because it finishes with an excessive battery-energy deficit. A rejection reason is stored when terminal compliance fails or a simulation raises an error.

12.6 Objective, ranking, and tie-break

$$\text{minimize } C_{km} = (C_{fuel} + C_{grid\text{-}equivalent\ electricity}) / d_{route} \tag{12.6}$$

Only feasible rows are sorted. Operating cost per kilometre is the primary objective; fuel consumption in litres per 100 km is the ascending tie-break. The optimization table displays the highest-ranked feasible rows, up to ten. Stable catalog IDs are retained so a result can be reproduced from the workbook.

12.7 What happens after the search

1. The status line reports the number of evaluated combinations; during the run it also shows the latest modeled cost.
2. All evaluated rows retain compatibility, feasibility, rejection reason, cost, energy, terminal SOE, unmet energy, and estimated-mass fields.
3. The Optimization Ranking tab receives the top feasible table, limited to ten rows.
4. The best feasible result becomes the app's current result, so the Signals, Detailed Plot, and KPI tabs update to that configuration.
5. Cancel requests are honored between candidate simulations; the already completed rows remain the meaningful partial search history in memory.

12.8 Worked ranking and responsible interpretation

The refreshed 144-case Battery 1 / Motor sweep retains the ten highest-ranked feasible rows. The leading row uses BAT-12 with MOT-12 at approximately 0.3130 EUR/km under the auxiliary-active-dump regeneration policy. Because all component values are synthetic, this ranking demonstrates workflow rather than a procurement recommendation.

Table 12.3. Top five feasible rows in the current 144-case study

Rank	Battery 1	Motor	Cost (EUR/km)	Electricity (kWh/km)
1	BAT-12	MOT-12	0.3130	1.7387
2	BAT-11	MOT-12	0.3150	1.7497
3	BAT-12	MOT-11	0.3164	1.7575
4	BAT-10	MOT-12	0.3170	1.7613
5	BAT-11	MOT-11	0.3183	1.7686

Correct interpretation
A first-ranked row is the best feasible row within the combinations that were actually evaluated, under the selected route, mass, prices, initial SOEs, component data, and terminal-energy rule. It is not automatically a production recommendation or a global optimum when the evaluation budget truncates the enabled search space.

13 Route Library and Provenance

13.1 Route categories

The database contains 20 routes: three official EU VECTO passenger missions, eight geographic long-distance coach corridors, and nine synthetic sensitivity profiles. The Route_Catalog records the region, source basis, organization, URL, version or retrieval time, licence, distance, duration, maximum speed, and notes.

13.2 VECTO missions

The Urban, Suburban, and Coach declaration-mode mission cycles are unmodified source files from the European Commission VECTO repository. The converter applies a deterministic bus driver with 1.0 m/s2 acceleration and 1.3 m/s2 braking bounds, honors declared stops, preserves the raw distance-domain trace, and produces a 1 Hz time history for this model.

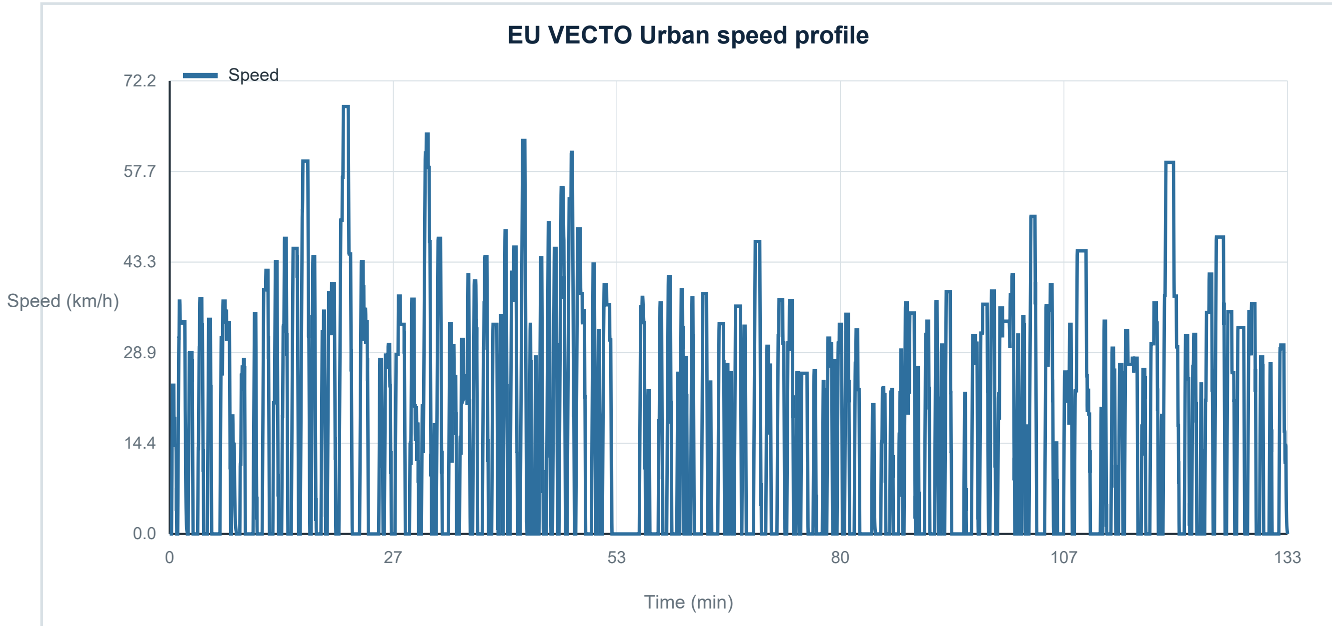

Figure 13.1. Default EU VECTO Urban mission after deterministic time conversion

13.3 Long European corridors

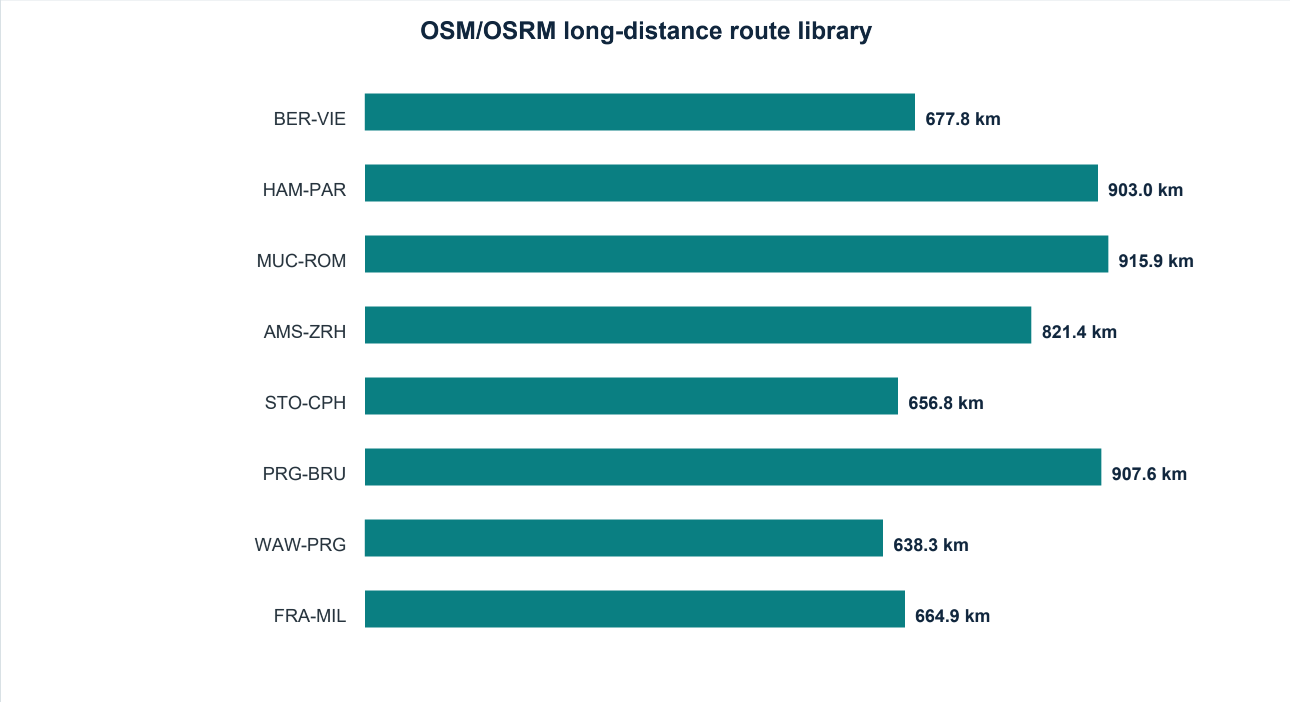

Figure 13.2. Geographic coach corridors in the requested 600-1000 km band

Table 13.1. Long-route source distances and adapted mission durations

Route	Countries	Distance	Adapted duration
Berlin, Germany to Vienna, Austria	Germany / Austria	677.8 km	8.45 h
Hamburg, Germany to Paris, France	Germany / France	903.0 km	11.18 h
Munich, Germany to Rome, Italy	Germany / Austria / Italy	915.9 km	11.54 h
Amsterdam, Netherlands to Zurich, Switzerland	Netherlands / Germany / Switzerland	821.4 km	9.73 h
Stockholm, Sweden to Copenhagen, Denmark	Sweden / Denmark	656.8 km	8.34 h
Prague, Czechia to Brussels, Belgium	Czechia / Germany / Belgium	907.6 km	11.12 h
Warsaw, Poland to Prague, Czechia	Poland / Czechia	638.3 km	8.32 h
Frankfurt, Germany to Milan, Italy	Germany / Switzerland / Italy	664.9 km	8.34 h

Route fidelity warning
OSRM supplies road-network distance and estimated segment durations but not elevation. Long-route speed is capped at 100 km/h, a 45-minute stationary break is inserted after each 4.5-hour driving block, and grade is set to zero. Mountain-route energy will therefore be understated until elevation is added.

14 Verification, Validation, and Test Evidence

14.1 Layered evidence

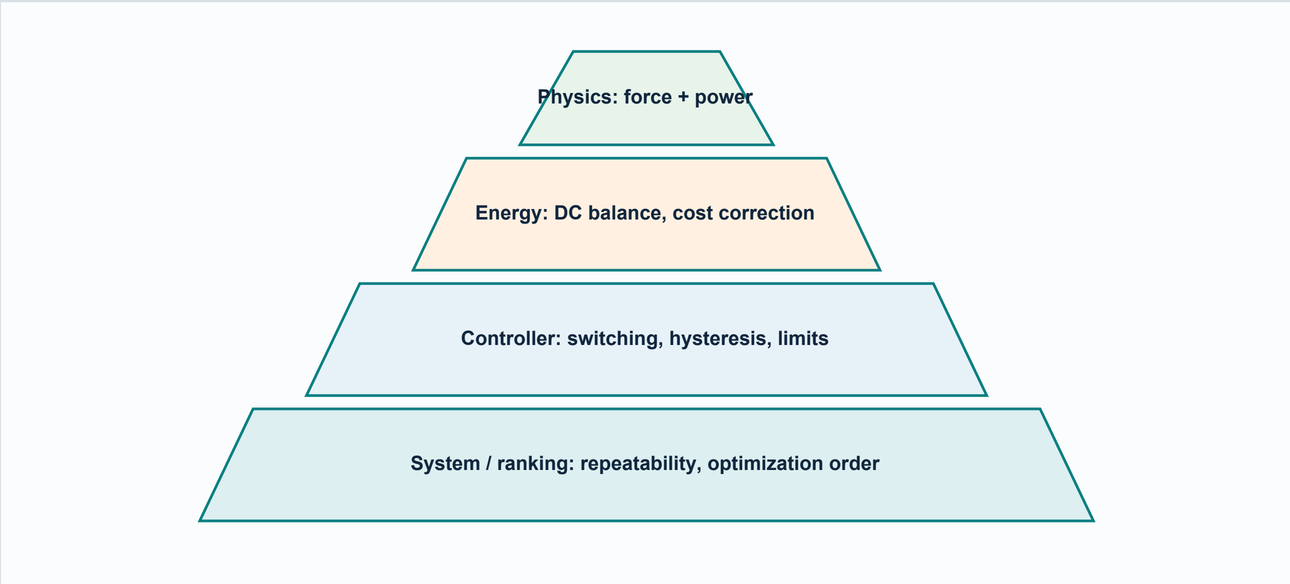

Figure 14.1. Verification pyramid used by the automated suite

The automated suite combines analytical comparisons, behavioral tests, traction-isolation tests, constant genset-power checks, regeneration-priority and full-active-battery load-bank checks, saturation tests, data-validation failures, conservation checks, repeatability, terminal-cost checks, charge-sustaining logic, and optimizer ranking. All twenty-two scenarios currently pass.

14.2 Test matrix

Table 14.1. Automated MATLAB test suite

Test	Purpose	Result
ConstantSpeedLevel	Constant-speed wheel power matches hand calculation	PASS
StationaryAuxiliary	Auxiliary load consumes battery energy at zero speed	PASS
LevelAcceleration	Acceleration increases wheel demand	PASS
PositiveGrade	Positive grade increases wheel power	PASS
DownhillRegeneration	Negative grade produces regenerative DC power	PASS
RegenerationPriority	Regeneration supplies auxiliaries, then the active battery, then the load bank	PASS
BatterySwitching	Low active pack deterministically switches to standby pack	PASS
StandbyGensetCharge	Genset charges only the standby battery	PASS
GensetTractionIsolation	Genset output never offsets traction-bus demand	PASS
GensetHysteresis	Genset starts for low standby SOE without chattering	PASS
BatteryUpperLimit	Battery energy never exceeds upper limit	PASS
BatteryLowerLimit	Battery energy never falls below lower limit	PASS
MotorSaturation	Motor saturation reports unmet wheel power	PASS
GensetBestEfficiencyPoint	Genset runs at one constant optimum power	PASS
RouteValidationFailure	Duplicate route time is rejected	PASS
IncompatibleSelection	Compatibility filter records rejection reason	PASS
EnergyConservation	Cumulative DC residual remains within tolerance	PASS
Repeatability	Identical inputs produce identical scalar KPIs	PASS
ZeroSpeedStability	Zero-speed calculation remains finite	PASS
TerminalCostCorrection	Grid-equivalent terminal correction matches hand calculation	PASS
ChargeSustainingComparison	Terminal combined-SOE tolerance is enforced	PASS
OptimizationRanking	Feasible configurations are sorted by total cost per kilometre	PASS

14.3 Independent hand check

For the current 19,000 kg default selection, the constant-speed level-road test compares the model wheel power against an independent analytical expression. Both calculations give 20.699 kW at the test condition. The test also confirms that the selected total mass enters the dynamics directly.

Passing tests do not validate synthetic data
Tests show that the implementation behaves consistently with its equations and stated limits. They do not prove that catalog efficiencies, BSFC values, component ratings, or route assumptions represent a production vehicle.

15 Worked Example: Default VECTO Urban Case

15.1 Case definition

The default case uses the VECTO Urban route, two BAT-12 packs, two MOT-12 rear hub motors, GEN-05, a 19,000 kg total vehicle mass, 85% initial Battery 1 SOE, 75% initial Battery 2 SOE, 1.65 EUR/L diesel, and 0.18 EUR/kWh electricity.

15.2 Signal interpretation

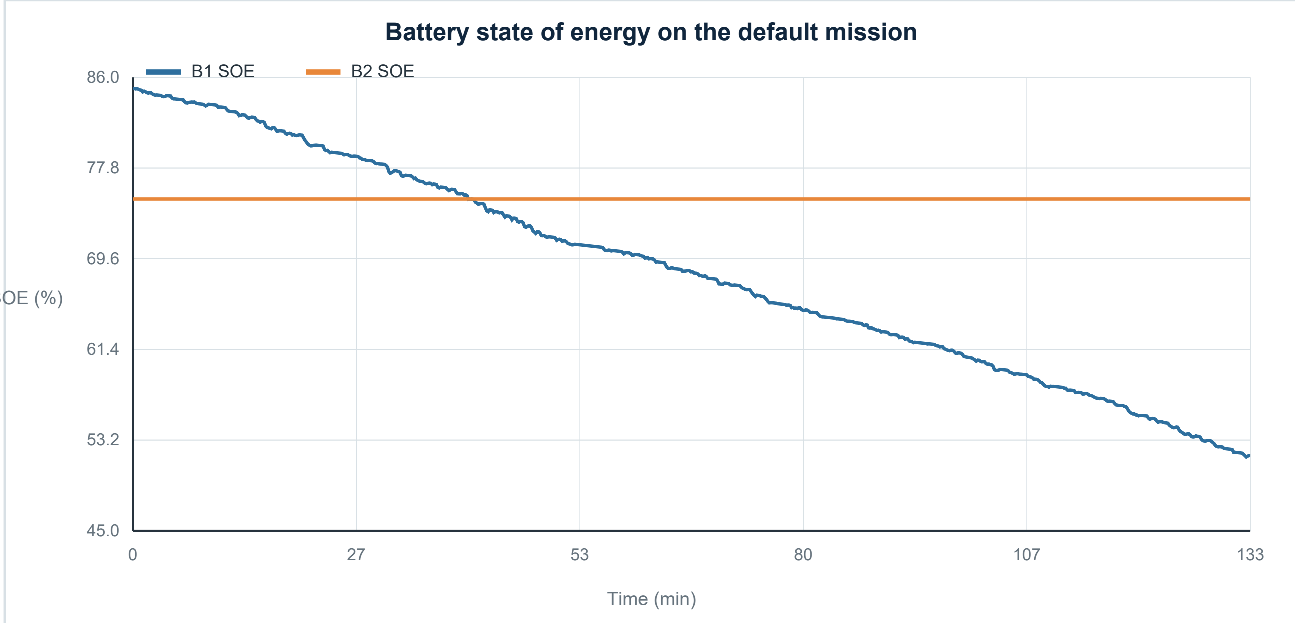

Figure 15.1. Default dual-battery SOE histories and active-pack behavior

Battery 1 is initially active and ends at 51.8% SOE; Battery 2 ends at 75.0% SOE. Regeneration is no longer redirected to the standby pack: 3.650 kWh supplies auxiliaries first, 21.934 kWh is accepted at the active-battery terminal, and 0.000 kWh is dissipated in the resistor bank. The genset does not start because the standby pack remains above the fixed 30% start condition.

15.3 Summary metrics

Table 15.1. Default-case results

Metric	Value	Interpretation
Route distance	39.787 km	distance integrated from the prepared 1 s trace
Grid-equivalent energy	69.176 kWh	terminal pack deficits divided by grid efficiency
Electrical intensity	1.7387 kWh/km	equivalent replenishment basis
Operating cost	0.3130 EUR/km	fuel plus grid electricity divided by distance
Available regenerated energy	25.584 kWh	non-negative energy available from negative motor DC power
Regeneration to auxiliaries	3.650 kWh	first-priority direct auxiliary supply
Regeneration to active battery	21.934 kWh	second-priority accepted terminal charge energy
Resistor-bank energy	0.000 kWh	third-priority controlled waste heat
Auxiliary energy	20.815 kWh	base and HVAC electrical demand
Unmet traction energy	0.000000 kWh	zero: case is feasible under the project criterion
Energy-balance error	0.000000000 kWh	zero within numerical reporting precision

Worked-example conclusion
The case is internally feasible and conservative under equivalent replenishment. It is not charge sustaining because combined terminal SOE is below the initial value, and it is not a real-vehicle performance claim.

16 Long-Route and Mass Feasibility Studies

16.1 Mass sweep

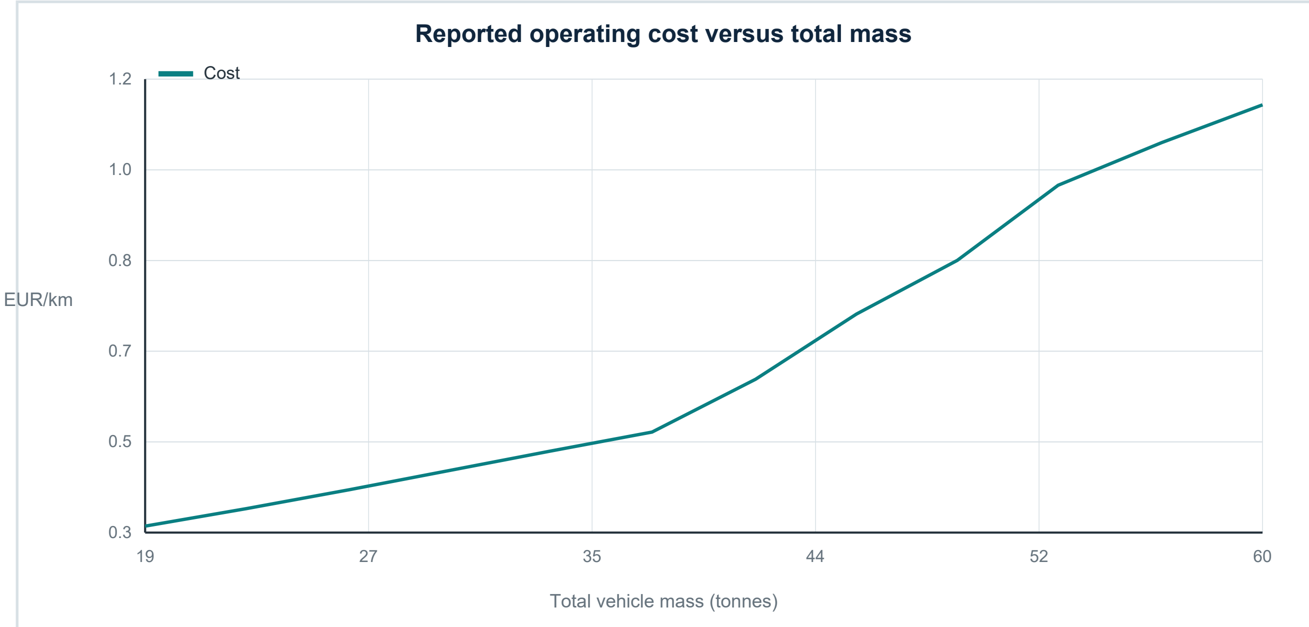

Figure 16.1. Cost trend must be read together with unmet traction energy

At 19 tonnes the default route has zero unmet traction energy. Small shortfalls begin near the next mass variant and reach 21.424 kWh at 60 tonnes. A cost trend alone therefore mixes feasible and infeasible cases; the feasibility flag is the governing result.

16.2 European long-route study

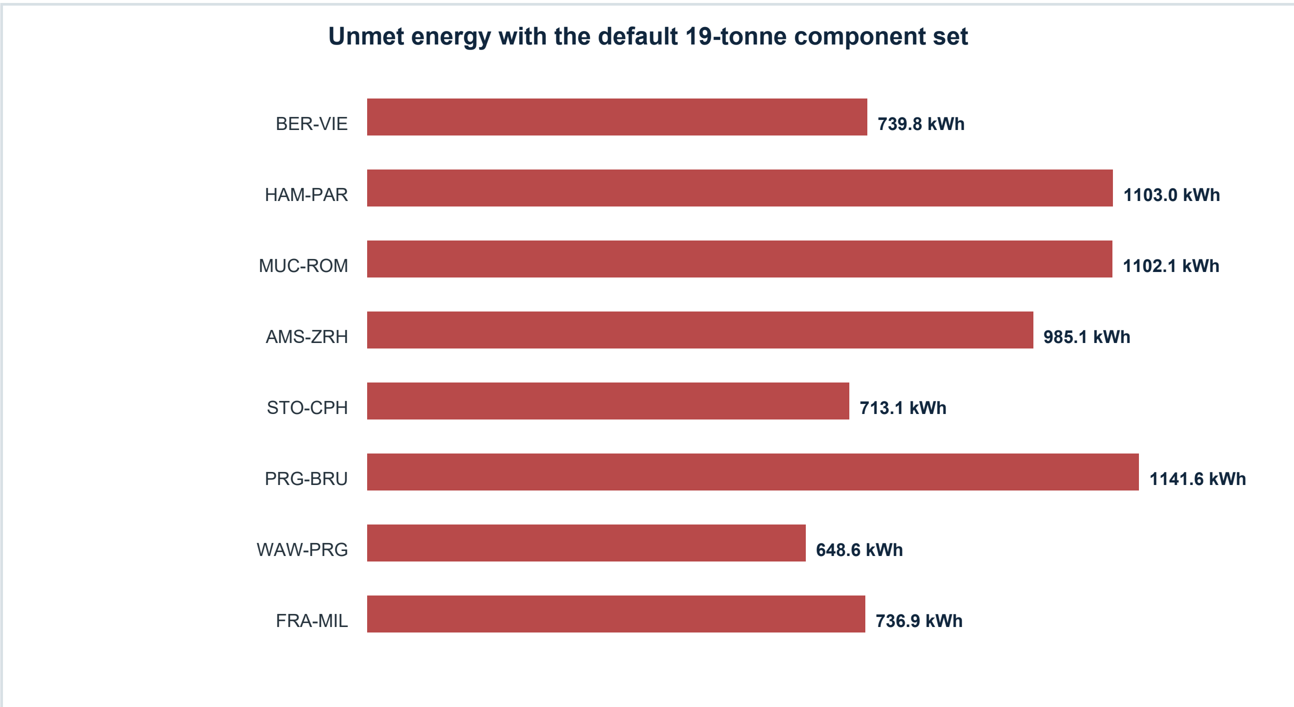

Figure 16.2. All long-route default cases are infeasible despite apparently low reported costs

The long missions integrate correctly, but the current default architecture cannot supply the complete requested traction/DC energy after stored energy is depleted. Unmet energy ranges from roughly 649 to 1,142 kWh. The low cost values in these runs are therefore not design candidates; they are examples of why feasibility must precede economics.

Table 16.1. Default-component long-route outcomes

Route	Simulated km	Fuel (L)	Unmet (kWh)	Feasible?
EUR-BER-VIE	677.4	37.8	739.8	No
EUR-HAM-PAR	902.2	37.8	1103.0	No
EUR-MUC-ROM	915.0	37.8	1102.1	No
EUR-AMS-ZRH	820.9	12.2	985.1	No
EUR-STO-CPH	656.3	22.9	713.1	No
EUR-PRG-BRU	906.8	33.0	1141.6	No
EUR-WAW-PRG	638.1	37.8	648.6	No
EUR-FRA-MIL	664.8	37.8	736.9	No

16.3 How to make a long-route study meaningful

1. Increase motor and active-pack capability, add sufficient battery energy, or schedule external charging; do not count genset output as traction power.
2. Introduce realistic elevation and road-grade data before comparing Alpine corridors.
3. Calibrate speed, dwell, passenger load, HVAC, traffic, and rest-stop assumptions with measured or operator data.
4. Run bounded optimization with feasibility constraints, then compare equivalent replenishment and charge-sustaining results.
5. Repeat under temperature, mass, headwind, and component-degradation scenarios.

17 Limitations, Responsible Use, and Extensions

17.1 Current model boundary

Table 17.1. Fidelity gaps and extension paths

Missing fidelity	Why it matters	Recommended extension
Battery voltage/current and temperature	Power and efficiency limits change with electrical and thermal state	Equivalent-circuit and thermal network with calibrated maps
Ageing	Usable energy, resistance, and cost evolve with throughput and temperature	Cycle/calendar degradation and life-cycle cost
Closed-loop driver	Backward demand may exceed trackable acceleration or actuator limits	Driver controller and forward vehicle dynamics
Road elevation for long routes	Mountain energy and regeneration are understated	Map-matched elevation with smoothing and grade validation
Engine transients and emissions	Fuel and pollutants depend on warm-up and aftertreatment	Dynamic engine/aftertreatment model and measured maps
Wheel slip and lateral dynamics	Traction limits and stability are not represented	Tyre-road force limits and vehicle dynamics
Supplier data	Synthetic catalogs cannot support procurement	Version-controlled, validated manufacturer maps

17.2 Recommended development roadmap

Figure 17.1. A disciplined progression from concept model to validated engineering platform

17.3 Graduate exercises

1. Derive an analytical constant-speed energy-per-kilometre expression and compare it with the kernel over several masses and headwinds.
2. Replace scalar motor efficiency with a two-dimensional torque-speed map and quantify the change in route energy.
3. Add elevation to one OSM/OSRM route, validate grade smoothing, and compare energy with the zero-grade baseline.
4. Design an alternative supervisor that allows controlled parallel discharge and define new safety/feasibility tests.
5. Formulate a multi-objective search for cost, fuel, battery throughput, and component mass; produce a Pareto front.
6. Create a charge-sustaining urban case by sizing battery energy and the fixed-point standby charger; keep the 30% role threshold unchanged.
7. Add a simple battery thermal state and demonstrate a cold-weather power-derating scenario.
8. Construct a requirements-to-test traceability matrix for every limit and KPI in the model.

Final engineering lesson
A useful model is not only a set of equations. It is an agreement among data definitions, physical assumptions, interfaces, numerical methods, controller logic, evidence, and interpretation. When any one of those changes, the others must be reviewed.

Appendix A Quick-Start Laboratory

Use the following sequence for a first supervised laboratory session.

1. Open MATLAB R2025a and change to the project folder.
2. Run startup_hybrid_bus and confirm that the project loads without startup issues.
3. Run Results = run_hybrid_bus_simulation; inspect Results.Summary and Results.Validation.
4. Open HybridBus_BackwardModel.slx and trace signals from route input to energy accounting.
5. Launch app = HybridBusApp; repeat the default case and inspect all plot tabs.
6. Change only one factor, such as mass or route, and explain every KPI change before running a second factor.
7. Run TestResults = run_all_hybrid_bus_tests; confirm 22 PASS results.

```
db = load_hybrid_bus_database("HybridBus_ComponentDatabase.xlsx");
report = validate_hybrid_bus_database(db);
assert(report.IsValid, strjoin(report.Errors,newline));

overrides = struct('SelectedRoute',"EUR-BER-VIE", ...
                   'SelectedMass',"MASS-01");
Results = run_hybrid_bus_simulation( ...
    "HybridBus_ComponentDatabase.xlsx",overrides,'SaveResults',false);
```

Appendix B Signal and Parameter Reference

Table B.1. Principal logged signals

Group	Signal	Unit	Meaning
Vehicle	Speed_m_s / Acceleration_m_s2	m/s, m/s2	prescribed speed and filtered acceleration
Wheel	Demand_kW / Delivered_kW	kW	required and limit-delivered wheel power
Wheel	UnmetTraction_kW / UnmetRegen_kW	kW	mechanical shortfalls
Motors	ElectricalPower_kW / MotorSpeed_rpm	kW, rpm	DC demand/regeneration and motor speed
Auxiliary	Power_kW	kW	base, HVAC, and route-multiplied load
Regeneration	Available_kW / ToAuxiliary_kW	kW	available braking energy and first-priority allocation
Regeneration	ToActiveBattery_kW / ResistorLoadBank_kW	kW	second- and third-priority allocations
Battery 1/2	Power_kW / Energy_kWh / SOE	kW, kWh, 1	pack source/sink and stored state
Genset	ElectricalPower_kW / FuelRate_L_s	kW, L/s	constant standby-charger power and fuel flow
Genset	ChargeDestinationBattery	integer	standby pack receiving genset power; zero when off
Controller	ActiveBattery / StandbyBattery / Mode	integer	supervisory role and discrete mode
Energy	BalanceResidual_kW	kW	power-conservation diagnostic
Energy	UnmetDCPower_kW / RejectedCharge_kW	kW	source/sink saturation accounting including load-bank power
Energy	RejectedGensetCharge_kW	kW	charger mismatch kept distinct from regenerative dissipation

Table B.2. Principal parameter groups

Area	Principal parameters	Workbook source
Vehicle	mass, gravity, Cd, frontal area, rolling resistance	Mass / Tyre / Vehicle
Route	time, speed, grade, dwell, auxiliary multiplier	Route sheets
Hub drive	wheel radius, ratio, efficiencies, torque/power/speed	Tyre / Final Drive / Motor
Battery	usable energy, SOE bounds, efficiencies, power, regen	Battery + Dashboard
Genset	optimum/max power, minimum times, BSFC, efficiency	Genset + maps
Supervisor	fixed 30% role threshold, standby upper-SOE target, auxiliary-active-dump priority	Architecture rule + control calibration
Load bank	regenerative surplus after active-pack limits	Derived power sink
Economics	fuel tank, prices, grid efficiency, terminal method	Dashboard / Vehicle / Prices / Optimization

References Sources and Project Files

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[4] Project OSRM API documentation, route service and annotation objects, https://project-osrm.org/docs/v5.24.0/api/.

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